INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

POLYURETHANE FOAM, POLYETHER TYPE, RIGID, FOR PACKAGING AND ENCAPSULATION OF ELECTRONIC COMPONENTS

This specification is approved for use by the US Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and tests for three types of two-component polyether type polyurethane foam, rigid, for packaging and encapsulation of electronic components.

1.2 Classification. The polyether type polyurethane foam will be furnished in the following types, as characterized by their densities and acquisition document (see 6.2):

Type I	-	2 pounds per cubic foot (lb/ft^3)
Type II	-	2 to 3 lb/ft^3
Type III	-	4 lb/ft^3

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL, 35898-5270 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) at the end of this document or by letter.

AMSC N/A FSC 5970

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-591	-	Boxes, Shipping, Fiberboard, Wood-Cleated
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-76	-	Tape, Packaging, Paper (For Carton Sealing)

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-F-83671	-	Foam-In-Place Packaging Materials, General Specification for

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-810	-	Environmental Test Methods and Engineering
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D149	-	Dielectric Breakdown Voltage and DielectricStrength of Solid Electrical Insulating Materials at Commercial Power Frequencies, Standard Test Method for
ASTM D150	-	A-C Loss Characteristics and Permittivity (Dielectric Constant) of Solid Electrical InsulatingMaterials, Standard Test Method for
ASTM D732	-	Shear Strength of Plastics by Punch Tool, StandardTest Method for
ASTM D1621	-	Compressive Properties of Rigid Cellular Plastics, Standard Test Method for
ASTM D1622	-	Apparent Density of Rigid Cellular Plastics, Standard Test Method for
ASTM D1623	-	Tensile and Tensile Adhesion Properties of Rigid Cellular Plastics, Standard Test Method for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-government standards and publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

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3.2 Material. The polyurethane foam compound shall consist of a polyether type base resin and a separate isocyanate-type base resin and a separate isocyanate-type curing agent, so formulated as to produce, upon mixing, a polyurethane foam of a specified density certified by the manufacturer (see 6.2).

3.3 Storage life. The foam shall conform to the requirements of this specification after a storage period which shall be not less than 6 months for type I and III and 24 months for type II when stored in the original unopened container at a temperature between 16 and 32° Celsius (C) [60 and 90° Fahrenheit (F)]. The storage life shall be certified by the manufacturer (see 6.2).

3.4 Physical properties. The physical properties of the cured material shall be in accordance with table I (see 4.7).

TABLE I. Physical properties

Property	Type I	Type II	Type III
Density, lb/ft^3 ----------	2 ± 10%	2 to 3	4 ± 10%
Tensile strength, psi, lb/in^2) minimum ----------	-	10 @ 25°C ± 1°(77°F ± 2°)	-
Compressive strength, psi, (lb) minimum ----------	-	3	-
Shear strength, psi, (lb) minimum -	15	6	15
Dielectric constant, maximum at 25°C(77°F) and 9375 MHz -------	1.25	-	1.25
Dielectric strength, V/mil, minimum ----------	-	45	-
Dimensional stability, % increase in volume between -54°C(-65°F) and 68°C(155°F) ----------	1.5	-	1.5
Flammability ----------	-	self extinguishing	-
Temperature resistance, 1 hour at 86.9 ± 3°C(190 ± 5°F) (loss of properties) ----------	-	None	-
Humidity growth, % increase in volume, % loss of physical property values ----------	-	10% 60%	-
Fungus resistance ----------	-	no fungus growth	-

3.5 Workmanship. The polyurethane foam shall meet all requirements of this specification. The materials shall be clean and free of foreign matter, gel particles and

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other contaminants that would affect the intended purpose. The foam shall be of uniform quality, condition and appearance (see 4.5.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet al requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection see (4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be conducted at a temperature of 16 to 32°C (60 to 90°F), relative humidity of ambient to 95 percent and barometric pressure of facility ground.

4.4 First article inspection. The first article inspection shall consist of the inspections and tests specified in table II in an order acceptable to the procuring activity. The first article sample shall consist of a one pint kit containing all required ingredients of each type of material, manufactured under the same conditions as those proposed for subsequent production. Quantity production shall be withheld until the first article sample has been determined to be satisfactory by the procuring activity.

4.5 Quality conformance inspection. Quality conformance inspections shall be as specified in table II.

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TABLE II. Quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph
Density	3.4, table I	4.7.2
Tensile strength	3.4, table I	4.7.3
Compression strength	3.4, table I	4.7.4
Shear strength	3.4, table I	4.7.5
Dielectric constant	3.4, table I	4.7.6
Dielectric strength	3.4, table I	4.7.7
Dimensional stability	3.4, table I	4.7.8
Flammability	3.4, table I	4.7.9
Temperature resistance	3.4, table I	4.7.10
Humidity growth	3.4, table I	4.7.11
Fungus resistance	3.4, table I	4.7.12
Workmanship	3.5	4.5.3

4.5.1 Sampling. For sampling purposes, a one pint kit containing all required ingredients shall be considered as a sampling unit. The number of sampling units shall be as specified by the procuring activity (see 6.2).

4.5.2 Lot formation. A lot shall consist of a batch of material manufactured in one continuous operation by the same process, by the same manufacturer, using the same batches of raw materials and submitted for 100 percent inspection at one time.

4.5.3 Examination. The polyether type polyurethane foam shall be examined for compliance with the requirements specified in 3.2 through 3.5.

4.6 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5. The inspection of marking for shipment and storage shall be in accordance with MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract (see 6.2).

4.7 Methods of inspection.

4.7.1 Test specimen preparation. Utilizing the samples selected as specified in 4.5.1, the test specimens shall be prepared as specified in the applicable test method.

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Five specimens shall be prepared for tensile strength, compression strength, shear strength and dimensional stability tests. One specimen shall be prepared for the remainder of the tests. The foam compound used for preparing specimens shall be mixed and cured in accordance with the manufacturer's recommendations. Specimens prepared for dimensional stability testing shall be four inches high and three inches in diameter. All other specimens shall be prepared in accordance with he applicable test method. No defects shall be allowed for dimensional stability or density tests.

4.7.2 Density. The material shall be tested in accordance with ASTM D1622 for conformance to table I.

4.7.3 Tensile strength. The material shall be tested in accordance with ASTM D1623 for conformance to table I.

4.7.4 Compression strength. The material shall be tested in accordance with ASTM D1621 for conformance to table I.

4.7.5 Shear strength. The material shall be tested in accordance with ASTM D732 for conformance to table I.

4.7.6 Dielectric constant. The material shall be tested in accordance with ASTM D150 for conformance to table I.

4.7.7 Dielectric strength. The material shall be tested in accordance with ASTM D149 for conformance to table I.

4.7.8 Dimensional stability. The material shall be exposed to seven heat cycles in accordane with figure 1 to determine conformance to table I.

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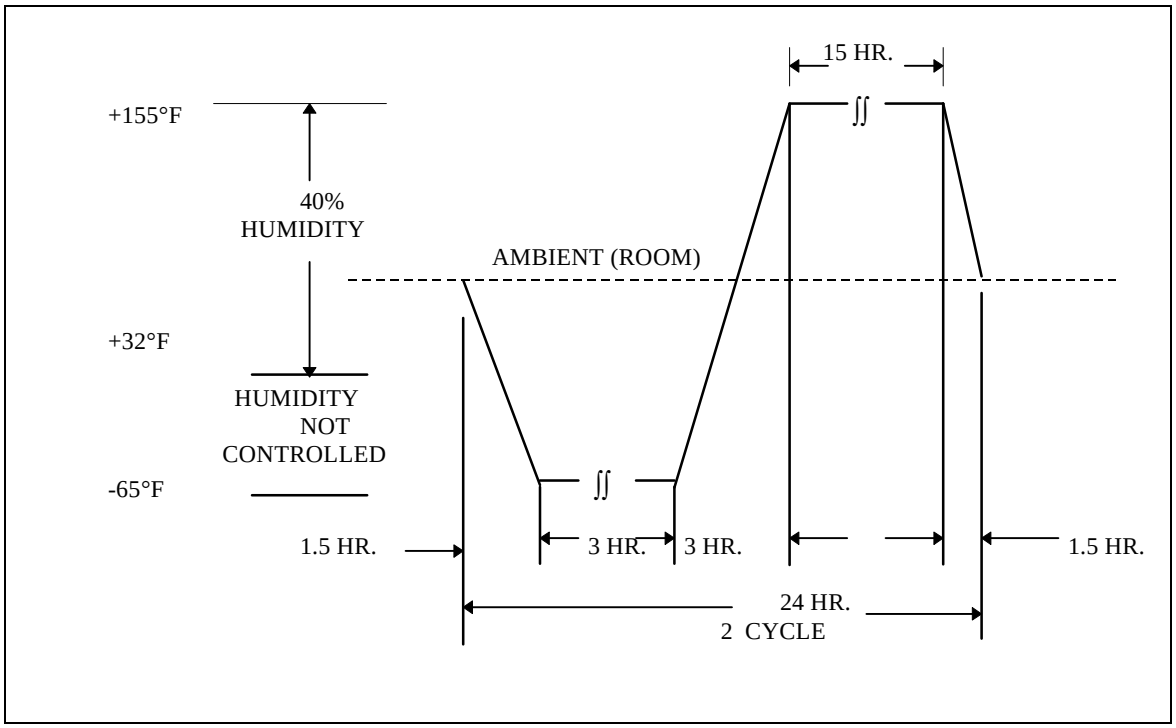


FIGURE 1. Heat cycle for dimensional stability test.

4.7.9 Flammability. The material shall be tested in accordance with MIL-F-83671 for conformance to table I.

4.7.10 Temperature resistance. The material shall be baked for a period which shall be not less than one hour at 86.9 ± 3°C(190 ± 5°F) for conformance to table I.

4.7.11 Humidity growth. The material shall be tested in accordance with MIL-STD-810, Method 507, for conformance to table I.

4.7.12 Fungus resistance. The material shall be tested in accordance with MIL-STD-810, Method 508, for conformance to table I.

5. PACKAGING

5.1 Preservation. Preservation shall be levels A, B or C (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packs. The polyether type polyurethane base resin and a separate isocyanate-type curing agent shall be unit packed in accordance with MIL-P-116, method IA-5. The polyether type polyurethane base resin and the isocyanate-type

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curing agent shall each be packed as one unit in a fiberboard box conforming to PPP-B-636.

5.1.1.2 Intermediate packs. Unit packs shall be placed in a close-fitting container conforming to PPP-B-636, type I, class 2 and closed as specified in 5.1.1.3.

5.1.1.3 Closure. Containers conforming to PPP-B-636, type I, class 2 shall be closed with six strips of 2-inch tape conforming to PPP-T-76. These strips shall be applied using three on the top seams and three on the bottom seams. When the bottom of the container is closed with staples or adhesive, only three pieces of tape on the top seams are required. The tape shall extend at least 3 inches onto each side panel. In instances where the end panel is less than 3 inches in depth, the tape shall extend to the full depth.

5.1.2 Level B.

5.1.2.1 Unit packs. Same as level A.

5.1.2.2 Intermediate packs. Unit packs shall be placed in close-fitting container conforming to PPP-B-636, type I, class 1 and closed as specified in 5.1.2.3.

5.1.2.3 Closure. Containers conforming to PPP-P-636, type I, class 1 shall be closed as specified in 5.1.1.3 using tape conforming to PPP-T-45, class 2 or 3.

5.1.3 Level C. Unit packs and intermediate packs shall be in accordance with MIL-STD-1190.

5.2 Packing. Packing shall be level A, B or C (see 6.2).

5.2.1 Level A. Units preserved as specified in 5.1 shall be packed in overseas-type exterior containers conforming to one of the following specifications: PPP-B-591, PPP-B-601 or PPP-B-621. The gross weight shall be not greater than 200 pounds. The container shall be strapped in accordance with the applicable container specification.

5.2.2 Level B. Units preserved as specified in 5.1 shall be packed in domestic-type exterior containers conforming to one of the following specifications: PPP-B-591, PPP-B-601 or PPP-B-621. The gross weight shall not be greater than 200 pounds.

5.2.3 Level C. Packing shall be in accordance with MIL-STD-1190.

5.3 Marking. Marking shall be in accordance with MIL-STD-129. Marking shall include but, not be limited to, the following:

a. Number of this specification.

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b. Type.

c. Manufacturer’s name and address.

d. Manufacturer’s designation.

e. Lot number or batch number.

f. Date of manufacture.

g. Shelf life.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The polyether type polyurethane foam is intended for potting and encapsulating electronic components to protect them from thermal and mechanical shock.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.2).

c. Whether first article sample is required (see 3.1 and 4.4).

d. Type required (see 1.2).

e. Requirement for certification of composition (see 3.2).

f. Requirements for certification of storage life (see 3.3).

g. Number of sampling units required for examination and testing (see 4.5.1).

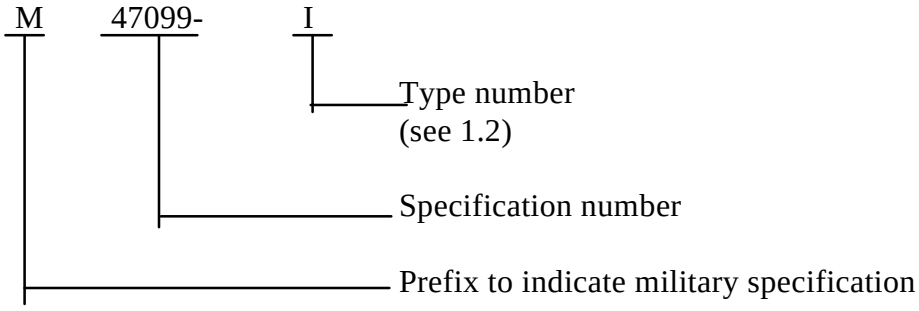
h. Level of preservation, packing and marking required (see 4.6, 5.1, 5.2 and 5.3).

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i. Quantity and size of containers.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidder should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Part or Identifying Number (PIN). The PIN to be used for polyurethane foam acquired to this specification is created as follows:



6.5 Metrication. Wherever inch/pound dimensions are used in this document, metric equivalents in accordance with FED-STD-376 will be acceptable.

6.6 Subject term (key word) listing.

Cellular plastic
Packing, chemical
Packing, insulation
Packing, plasticizer
Packing, resin base

6.7 Changes from previous issue. Marginal notations not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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